



Improving employability skills through non-placement work-integrated learning in chemical and food engineering: A case study

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ABSTRACT

Preparing work-ready chemical engineering graduates is achieved by integrating the technical skills and knowledge learned at university with employability skills required by industry. While this is most often made through industry placements, non-placement forms of work-integrated learning (WIL) can be highly effective in preparing graduates for the workplace without the issues of locating work placements and ensuring their quality. In this paper, the authors focus on a chemical engineering course that combines non-placement WIL with a problem-oriented/project-based learning methodology, and a problem-solving tool, the Integrated Product and Process Design (IPPD) framework. The authors present qualitative data from students, lecturers, and industry partners to evaluate whether the employability skills of creativity and teamwork are developed in the course. Through a process of qualitative analysis, the authors developed five key themes that provide a focused understanding of how the parts of the course relate to one another and drive student learning. The findings of this study indicate that the model of non-placement WIL evaluated was effective in building the defined employability skills; however, there are opportunities for iterative enhancement. The key learnings from this study may guide others interested in building non-placement WIL into chemical engineering education.

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1. Introduction

The development of competitive, work-ready graduates for the 21st century requires significant changes in the engineering curriculum towards a new model of “dynamic, hands-on learning and projects” (Accreditation Board for Engineering and Technology, 2017, p. 10) to replace the old style of “teach-memorize-test-repeat,” (Accreditation Board for Engineering and Technology, 2017, p. 10). To this end, the school of engineering at a private university located in Colombia is engaged in the process of curricular reform to transform the way engineering classes are designed and taught. This involved a move away from traditional approaches and towards a focus on preparing students to enter the engineering profession through problem-oriented/project-based learning (PO/PBL) methodologies (Ballesteros et al., 2019; Salcedo Galán et al., 2018).

The selection of a PO/PBL approach to building student employability skills in real-world conditions is a learning design method that many universities have taken (Griffiths et al., 2018). However, the nature of employability skills varies from discipline to discipline, and there is “a lack of consensus as to what constitutes employability skills and how they are levelled” (Griffiths et al., 2018, p. 891). This has contributed to a lack of clarity around ways for teaching and assessing employability skills and difficulties in their evaluation (Ajjawi et al., 2020; Jackson, 2015) which is compounded by a lack of research in this area as compared to research on methods of assessment of academic skills and knowledge (Ramberg et al., 2019).

Acknowledging the difficulties of assessing employability skills, in this paper the authors report on a study that evaluated the perceptions of students, lecturers and industry partners on whether the employability skills of creativity and teamwork, which were intentionally integrated but not formally assessed, were developed in a chemical engineering course that combined non-placement work-integrated learning (WIL) (Cooper et al., 2011) with an industry-based project. The authors present five key themes that

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were drawn from the data and an integrated course model that shows the relationships between the components of the course. The findings of this study indicate that the model of non-placement WIL used in the course was effective in building the defined employability skills; however, there are opportunities for iterative enhancement. The key learnings from this study may guide others interested in building non-placement WIL into chemical engineering education.

2. Background

2.1. Work- integrated learning

Work-integrated learning is an umbrella term that describes various practices across multiple disciplines, including placements, practicums, action learning, apprenticeships, service learning, and problem/project-based learning (Kaider et al., 2017). The main benefit of WIL is the role it plays in “developing work-readiness to the standard which industry expects of new graduates” (Jackson, 2015, p. 1). Consequently, WIL is recognized as a “key pedagogical strategy” (Stirling et al., 2016, p. 4) in preparing chemical engineering graduates for employment.

In some countries, WIL is mandated into “curricula and qualifications in order to promote student career development” (Govender and Wait, 2017, p. 1). In Colombia, where this study was based, WIL is not a mandatory component of chemical engineering programs. Only 18 % of Colombian undergraduate engineering students engage in the professional practice of some sort, although 50 % of Colombian chemical engineering students indicate a desire to engage in work-integrated learning during their undergraduate degrees (Cruz and Pineda, 2018).

Project-oriented/problem-based learning (PO/PBL) is a non-placement form of WIL (Jackson, 2017) that has been successful in engineering education (Ballesteros et al., 2019; Coronella, 2006; Simic et al., 2016). While project-based learning does not necessarily involve industry partnerships, it increases proximity to the workplace on top of the layer of authenticity and relevance of the problem-solving activity (Cooper et al., 2011). Project-based learning, in conjunction with industry partners, has many benefits over internships, which are the most utilized form of WIL in Colombia. The project-based model of non-placement WIL provides lecturers more control over the quality of student learning as compared to the highly variable experiences that students have through internships (Mutereko and Wedekind, 2016). Additionally, insufficient placements for all is a problem with internships, as is inequality in placements due to factors of race and social status (Mackaway et al., 2014; Tran and Soejatminah, 2017). Difficulties in maintaining high quality and consistent student experience across industrial contexts (Jackson, 2015), and differing expectations of the supervisory role (Theresa et al., 2016) are also problems that are mitigated by the PO/PBL approach to WIL.

2.2. Employability skills

The integration of employability skills in higher education courses is aimed at producing work-ready graduates (Billett, 2011). A large-scale international study, *The Global Skills Gap in the 21 st Century* (Quacquarelli Symonds, 2018), reports that the skills most relevant to employers globally are problem-solving, teamwork, and communication. That report describes a gap between the skills that graduates have and those that employers expect graduates to have when they enter the workforce. In Latin American countries, employers expressed a high level of satisfaction with “the technical skills of the graduates they employ” (Quacquarelli Symonds, 2018, p. 12), but not with their employability skills. The report indicates

that Latin American universities “do not necessarily provide enough opportunities for students to develop skills critical for the labor market” (Quacquarelli Symonds, 2018, p. 5). That is, the traditional teaching model focused on knowledge acquisition that is prevalent in engineering education in Latin American countries is not preparing graduates with the skills they need to transition smoothly into the workforce.

Creativity and innovation are essential employability skills for chemical engineers (Fernandez Rivas et al., 2020). However, there are challenges of defining and assessing the employability skill of creativity given that “creativity is complex and multifaceted in nature, [and] there is no single, universally accepted definition” (Treffinger et al., 2002, p. 7). Indeed, the expression of creativity is aligned with its disciplinary context (Fautley, 2018). In engineering, one definition of a creative thinker is someone who “while demonstrating a solid knowledge of the parameters of the domain in the highest levels of performance, pushes him or herself beyond those limits by means of new, unique or atypical combinations; discovering or critically perceiving new synthesis, and using or recognizing risk-taking to achieve a creative solution” (Lopez-Malo et al., 2016, p. 3). This aligns with the understanding by the lecturers of the course discussed in this paper, that creativity is the ability of students to design processes or products that show adaptability, functionality, and disrupt current models or ways of thinking or doing.

Teamwork is another important skill required of graduate engineers, and while it is difficult to define, it is regarded here as “a social strategy built upon knowledge, attitudes, skills, and the ability to combine cognitive appreciation from all team members” (Jorgensen et al., 2019, p. 2). High performing teams, such as Formula 1 pit-stop mechanics, are often seen as an ideal example of teamwork in engineering. Every individual has a defined task, and the individual’s goal is to carry out the job effectively, focusing primarily on time. However, this simplistic view of teamwork as establishing goals, planning tasks, and meeting objectives is not adequate when dealing with the industrial world and its dynamics. Team members often have multiple projects, which translate to numerous teams, and teams are not necessarily built before the tasks. Instead, multifunctional, multidisciplinary, and versatile engineers perform in a wide variety of environments; and more importantly, the tasks or projects are rarely repetitive and under an anticipated set of conditions.

2.3. Integrated product and process design

The Integrated Product and Process Design (IPPD) problem-solving tool (Alvarez, 2017) (Fig. 1) consolidates the different elements associated with the design of a product, its properties, its process, entrepreneurship, and innovation. The IPPD was the pedagogical means for promoting creativity and teamwork in the course discussed in this paper. It provided a logical and reasoned approach to solving problems, which is regarded as one aspect of creativity (Lopez-Malo et al., 2016; Schmidt and Charney, 2018). The IPPD tool was also used to support the development of teamwork skills in a wide variety of controlled and uncontrolled scenarios where the “learned practice” (Jorgensen et al., 2019) of teamwork and its components such as positive interdependence, individual accountability, interpersonal skills, and group processing could be built (Roger and Johnson, 1994).

The problem-solving methodology of the IPPD tool comprises three main parts: C: which refers to the process of identifying opportunities and generating ideas, and hereafter as *Conception*; I: *Innovation*; and E: *Entrepreneurship*. Conception involves the identification of opportunities and the ideation process using any known methodology such as brainstorming, systematic inventive thinking, or design thinking. Innovation involves

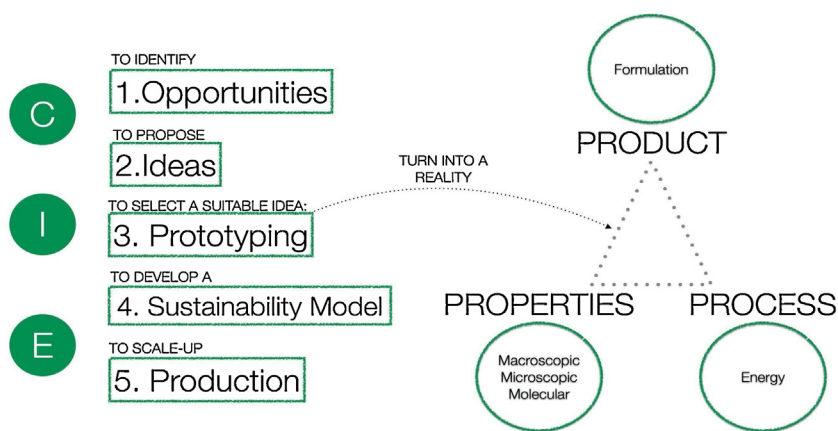


Fig. 1. The Integrated Product and Process Design, a design tool implemented by the Department of Chemical and Food Engineering at Universidad de Los Andes (Alvarez, 2017).

fast prototyping of suitable ideas using a multiscale approach (Cussler et al., 2012). Lastly, entrepreneurship requires the development of the business model and the design of the production plant.

The use of IPPD in the chemical engineering course reflects a philosophical approach that responds to the physical and political context in which the chemical engineering program has been developed. The location of the university in Bogotá, high in the Andes ranges and far from the nearest port, has meant that the focus for chemical and food engineering graduates is on the process and product innovation rather than on product production, as may be the case for chemical engineers in other contexts. Thus, creativity (leading to innovation) is a vital employability skill in this context, along with resourcefulness, motivation, adaptation, and improvisation, particularly when time and resource allocation constraints are present. In Bogotá, innovation in food and beverage, cosmetics, and specialty industries provide the predominant markets for chemical and food engineering graduates, and these industries are the focus for the problem-solving and industry partnerships in the course discussed in this paper.

2.4. The course

The *Chemical Engineering Challenge* (the course) is a non-placement-based WIL elective course of eight weeks (a full semester consists of 16 weeks). Departmental guidelines specify that electives have no more than 20 students. The course provides students with the opportunity to develop employability skills while working in teams to address a current chemical or food engineering opportunity (understood here as a problem, challenge, new line of business, etc.) being experienced by an industry partner. In the course, students need to solve unique industry challenges that require a specific set of skills to be mobilized for its resolution, for example, the conception of a new product. The course has the benefit of engaging students with industry partners, but, unlike the program capstone project, it is not fully aligned with the suite of ABET (Accreditation Board for Engineering and Technology) student outcomes.

The transition towards a PO/PBL model in the school of engineering was motivated by the requirements for extending the international accreditation with ABET. In this study, with its focus on employability skills, the authors note that the course is aligned with several ABET student outcomes (1, 2, 3, 4, and 5), noting that emphasis on teamwork was made only in one in particular: “an ability to function effectively as a member of a team” (Accreditation Board for Engineering and Technology, 2019). Although creativity

is not an employability skill identified by ABET, the Department of Chemical and Food Engineering (the department) has identified it as an essential skill for effective problem-solving in the chemical engineering profession.

Working on authentic industry problems heightens the relevance of learning by “exposing students earlier to real-world challenges [which] is increasingly important as society confronts the demanding population and infrastructure challenges of the next several decades” (Michigan, 2017, para. 13). With the support of their two lecturers (who team-taught the course) and industry partners, students work in small teams to creatively solve the authentic problem while also developing project management, communication, resourcefulness, conflict resolution, and other employability skills.

The course has been carefully structured. Orientation to the course takes place in four stages. First, lecturers meet with the participating industry partners to establish the main points of the challenge, such as technical aspects and expectations, and to schedule follow-up meetings with the students. The authentic industry problems on which the course is based are discussed between the industry partner and the course lecturers prior to being presented to students to ensure that the problems are suitable. That is, they must be doable within the semester time frame, they must include experimental work, and the projects need to be implementable by the industry partner. In addition, there must be no intellectual property restrictions associated with the project, and there must be some prior work related to the challenge that demonstrates that the project can be done and that at least one solution is possible. Examples of authentic problems that student teams worked on during the first semester of 2019 include the following: (1) Develop a viable process through which cacao pulp can be transported by plane from Colombia to Japan, without additives that could change its flavor profile; (2) Reformulate the process through which a certain brand of cosmetics is made, keeping the same ingredients but aiming to reduce energy consumption, and (3) Develop a new product using by-products obtained from brewing processes. In addition, the industry partners also allocated a consultant (an employee) who liaised directly with the students. Each consultant met with students for 20 min every week to respond to inquiries made by the students and sent to the partner three working days beforehand.

Second, teams of four or five students are established using a personality diversity indicator test based on the Birkman method (Birkman Fink and Capparelli, 2013).

Third, the challenges are presented to the students by the lecturers. Through a self-regulated process, each team then chooses the challenge it will confront.

Finally, the lecturers communicate the conditions and deliverables of the projects to the teams. The course deliverables include a project charter (Rueker and Radzikowska, 2008), a Pugh matrix to show the criteria-based decision making used to select a solution (Pugh, 1981), a Lean CANVAS to present a snapshot of the proposed solution and its fundamental assumptions (Osterwalder and Pigneur, 2010), a Gantt chart to scope out the project actions and timeframe (Wilson, 2003), and a RASCI matrix to clarify the roles and responsibilities of team members (Cabanillas et al., 2012; Olander and Landin, 2005). Also, each team makes two presentations to their peers before the final pitch delivery of their solution to their industry partner, who is involved in the formal assessment of the project solution.

The course is laboratory oriented, and students develop and test their ideas in fully-equipped laboratories that are equipped to handle chemical and biological waste. Every prototype the students create is required to be scientifically sound, and the prototypes are discarded based on properties measured in the labs. The course is also theoretically grounded, and students participate in some theoretical classes (5 out of 16 possible classes over the 8-week course) where the lecturers introduce project management tools and explain what resources are available for the teams to use while developing a solution to the challenge.

Every week during the course, the teams meet with a representative from their industry partner and with their lecturers to discuss technical details, future work, problems, and solutions. For each deliverable, feedback is given by the lecturers during the weekly sessions with each team. At the end of the course, the teams present their solutions to their industry partner at events hosted at their headquarters. The final grade for each project takes into account the project deliverables, as well as the observations and feedback of the industry partner and the lecturers. Although creativity and teamwork are integral parts of the IPPD problem-solving process, there is no explicit assessment of the employability skills that are implicitly built or strengthened during the course.

The authors note that the evaluation presented in this paper is of the first iteration of the course; however, its design was modeled on several other courses previously taught in the department, which used the PO/PBL methodology. This course was the first one that combined industry partners with the PO/PBL model, but not the first that encouraged creativity and innovation. The PO/PBL approach draws on the experience that the department cultivated over the five years prior to the implementation of the course, and the evaluation focuses specifically on the extent to which the employability skills of creativity and teamwork are developed through the integration of industry contact and authentic industry problems.

3. Research design

3.1. Research question

The focus of this study was to investigate the following research question in the context of the non-placement work-integrated learning course, the *Chemical Engineering Challenge*: Are the employability skills of creativity and teamwork developed in the course?

3.2. Research aims

This study aimed to evaluate the development of the employability skills of creativity and teamwork that were designed into the course, but which were not explicitly assessed.

The authors also wanted to collect evidence of alignment (or not) of the course with the ABET outcome: “an ability to function

effectively as a member of a technical team” (Accreditation Board for Engineering and Technology, 2019).

An additional aim was to develop an evidence base for the iterative enhancement of the course.

3.3. Participants

All students enrolled in the course in the first semester of 2019 were invited to participate in this study voluntarily. Out of the class, thirteen students (62 %) accepted. Of the participants, seven were male (54 %) and six (46 %) female. The participants were aged between 19 and 22 years old. All participants were in the latter stages of their engineering degree, in the sixth to eighth semester. Only three of the participants (23 %) indicated that they had any prior work experience before enrolling in the course. One industry partner (working with two teams) agreed to participate in this study out of three industry partners involved in the course. The two lecturers who team-taught the course participated as teacher-researchers in this study.

3.4. Data collection and analysis

Qualitative data were collected through a survey and focus groups with student participants, through semi-structured interviews with industry partners and through the written reflections of lecturers. All data were collected in Spanish, later transcribed and translated by the authors into English. Data saturation was achieved with students through the surveys and focus groups, indicating the data’s robustness and reliability. The case-based nature of the study produced rich context-specific data that is not intended to be generalizable. However, we note that the key findings may be relevant for consideration in other non-placement chemical engineering courses.

The survey compiled anonymous individual responses from students about key aspects of the course through open-ended questions. Three focus groups were conducted with four to five students in each. The focus groups were facilitated by the two researchers/authors who were not lecturers in the course to allow students to speak openly. The semi-structured interview was conducted with the industry participant at their premises. As with all semi-structured interviews, new topics and questions emerged during the process (Green et al., 2007). Triangulation of data from all sources provided a comprehensive picture of the course. An overview of the questions asked of participants is shown in Table 1.

While there are an array of methods for analyzing qualitative data (Leech and Onwuegbuzie, 2007), the approach used in this study was thematic analysis (Bazeley, 2009). This took place through a rigorous and systematic process over four stages: data immersion, coding, creating categories and identifying themes (Green et al., 2007). While appearing linear and sequential, the stages of analysis occurred concurrently. From the start, the authors were immersed in the data collection and its transcription, taking notes, and identifying preliminary codes, categories and themes. Repeated listening to audio recordings of the interviews and focus groups and rereading of the written texts from those, as well as the survey responses and lecturer reflections, deepened their knowledge of what was said and stimulated the analytical process “where one begins to ‘incubate’ ideas about the possibilities of analysis” (Green et al., 2007, p. 547). Two of the authors individually coded the data manually, color coding blocks of text, and writing notes about key concepts emerging. The process started with some predefined codes relating to creativity and teamwork, while other codes were drawn inductively from the data. Throughout the process, a constant discussion took place about the relationships between the data, which led to the working and reworking of codes into “coherent categories” (Green et al., 2007, p. 248).

Table 1
Summary of questions asked, by research method and participant group.

Method	Participant	Summary of guiding questions or prompts
Online survey	Students	Demographic data: Semester of study, gender, age, prior work experience • How did you and your teammates organize the information and course activities to establish the different opportunities to solve the challenge? • In your own words, how would you define innovation? • In your own words, in what ways was your final product innovative? • How did you and your teammates resolve the conflicts that may have occurred while working together? • What external factors influenced your work as a team? • Describe your overall experience in this course.
Focus groups	Students	<i>Creativity:</i> • In what ways did the IPPD process serve as a tool for your professional development as a chemical engineer? • Describe the process of analyzing the criteria and selecting opportunities to solve the challenge as a team. • In your own words, in what ways was your final product innovative? <i>Teamwork:</i> • How were tasks and roles decided on the team? • Explain how the RASCI matrix was used (if so) to define the different roles. • Do you think the conflict resolution matrix helped when conflicts occurred? How? Why? • In what ways was the IPPD process helpful when creating the final product? • Describe the communication between team members (positive aspects and areas for improvement) • What external elements influenced the organization and performance of the team? • Did the weekly meetings with the professors influence the work? If so, describe how. • How did you communicate with the industry partners?
Semi-structured interviews	Industry partners	• What were your expectations for this challenge? • Why were you interested in participating in this challenge with the University? • From an industry perspective, what is understood by innovation and creativity? • In your opinion, did the students exhibit the necessary skills to solve the challenge? • What were your initial expectations of the course? • How would you describe the teams' performance? (organization, communication, time-management, expectations) • How did you perceive the student-student interaction? • Did you notice conflict within the teams? • How did students organize themselves? • Would you describe the final products as innovative? Why or why not? • Describe the communication process with students. • How was the interaction with the students?
Written reflections	Lecturers	• General perceptions about the course • Learning opportunities • Greatest challenges

The other two authors were brought into the analytical process to review the codes and the logic of their categorization, which led to their further refinement. Jointly, all of the authors examined the categories for their relevance to the research questions and their significance in explaining and interpreting the nature of the development of the employability skills of creativity and teamwork. Descriptions of the themes were reworked extensively as the authors went back to the data and the literature to make sense of the research findings in the context of the course, the educational theory and practice. Through the analytical process, the authors described the themes as well as challenged and linked them (Bazeley, 2009). This systematic process led to the development of a model of non-placement work-integrated learning that shows the integration of the themes in the context of the course structure and procedures (see Fig. 2).

Ethical approval for this study was obtained from the ethics committee of the School of Engineering, Universidad de Los Andes, Colombia.

4. Results

Analysis of the data led to the identification of five themes that encapsulate the findings of the study. In this section, the authors present a description of each of the themes and supporting data to illustrate each theme.

4.1. Theme 1: Engagement in problem-solving of authentic industry problems develops students' professional identity and motivation

This theme reflects that the model of non-placement work-integrated learning used in this course, whereby students have the opportunity to solve authentic problems posed by industry, contributes to students' building employability skills and developing a professional identity. The process of engagement with industry, while they are developing a solution to an industry problem and with the prospect that industry will implement their solution, is highly motivating for students.

The data revealed that the integration of disciplinary knowledge with work-based problems linked students to the real work of a chemical engineer. This was evident from students' comments, such as: "The idea of solving real industry challenges allowed me to put into practice my knowledge as an engineer." Students found the authenticity of the problems they were involved in solving and the proximity they had to the industry as "enriching," and this was reinforced through industry visits. For students, these visits were "a learning opportunity to go to the plant and to get to know the company and, by doing so, to deepen their understanding of the context of the industry problem. The direct contact students had with an industry partner in the course was essential to the development of their professional identities, and students "began to visualize the possible challenges of [their] professional lives."

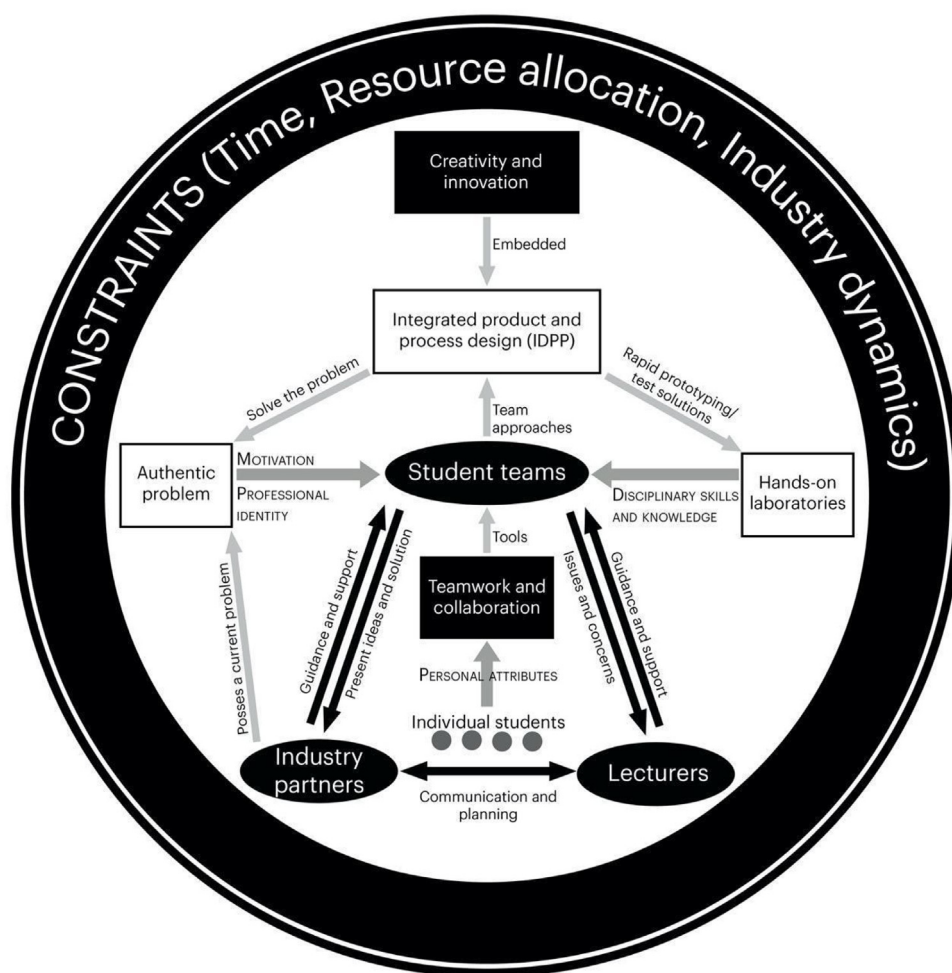


Fig. 2. The course model.

The course model also drove intrinsic motivation, with students more involved in the challenges of their projects than concerned about marks. For many students, this “was the first course where [they] cared more about the image of [the] product or the image of the university than the grade itself.” In the view of one of the lecturers, student motivation came from being asked to step up and perform as employees and live up to the expectations of the industry partner. The lecturer stated that “the course can enrich student learning because they experience real pressure from the workplace. It is not because the challenges are real. When the company supports students and treats them as employees, they feel the need to respond in the same way”. Students were also motivated by recognition from the industry partner, particularly when it was a well-known organization, as well as by the knowledge that their solutions could be implemented by industry.

4.2. Theme 2: An industry-based project model of work-integrated learning provides strong opportunities for students to develop employability skills in a structured and supportive environment

This theme reflects that the IPPD tool combined with scaffolding and weekly meetings with lecturers and regular engagement with and feedback from industry partners constructs an environment where the employability skills of creativity and teamwork are nurtured.

The nature of support from lecturers during the course was process-based, enabling teams to “reach a solution themselves, meeting the requirements of each case.” The support provided by lecturers and industry enabled teams “to gain a clearer idea of what [they] were doing. . . which in turn resulted in [them] putting more effort into doing a good project”.

The guiding role of lecturers took on particular importance when issues arose, such as when there was less contact between a project-group and industry partner than was optimal. As one student noted, “although we were never denied that help, on some occasions, it was difficult to obtain it.” Despite the importance of the industry partners role in the course, this comment reflects the inconsistencies in the commitment that occur from one industry partner to the next. The data indicate that when these inconsistencies occur, the lecturers take on a stronger role in supporting student teams, and mediate between the student teams and the industry partner.

4.3. Theme 3: Differing stakeholder expectations around timeframes, scheduling, and preparedness reflect the tensions that exist between the academy and the workplace, with the course providing a transition point between them

This theme encapsulates the conflicting priorities around the time that exists between students, academics, and industry and the tensions related to the time that needs to be navigated and negotiated for successful project outcomes to be achieved.

The data showed that the students, lecturers, and the industry partners believed that the eight-week timeframe for completing the project was short. Students understood that “the idea is that as an engineer, we need to provide solutions quickly” and for the industry, students needed to recognize that the short time frame mirrored the rapid pace of innovation in the workplace, where there is the need for the industry “to respond quickly to the competition and the market.”

However, the tight timeframe for completing the project was compounded by academic workloads in other subjects, which impacted how much time they could dedicate to the project: “You have to focus on other things, on other projects, on other exams. The course is important, but there is no time”. Students also indicated that the scheduling of the course in the second half of the semester compounded the short project timeframe, as that was a time of increased academic workload with other assignments falling due and exams scheduled. Students felt that the course would be better run in the first half of the semester or as a full semester course over 16 weeks.

For many students, the autonomy to choose their project, define its direction, and make rapid decisions was a new and “very challenging” experience. Students were “used to developing projects with an idea already proposed and with more time” and felt that they needed more time to do the project. In contrast, the industry partner thought students needed to be better prepared for the course. Still, they noted that “as similar challenges are presented, the skills and competencies required can be developed.” That is, by engaging in the challenge, students would learn and be able to “generate faster responses to [subsequent] situations.” As noted by one of the lecturers, in the course, “students can formally experience the types of challenges that the industry faces typically. . . [through a] student-in-training scenario.”

4.4. Theme 4: The links between problem-solving and creativity and innovation are poorly understood

This theme reflects the diverse understandings of creativity and innovation that emerged from the data. There was considerable variability in student responses around whether their project solution or the process they engaged was creative, though lecturers and the industry partner were clear that the teams produced innovative solutions to the challenges they addressed. As noted by one of the lecturers: “each team received a challenge, and each team delivered a solution that each of the companies can implement.” The variability in student perceptions of their creativity, or lack of it, is related to lack of a shared or consistent understanding of creativity as a component of innovation in the disciplinary context and contrasts with lecturers’ opinions of the centrality of creativity to innovation and the IPPD process.

For the lecturers, the successful resolution of an industry problem was regarded as a by-product of student creativity and innovation, whereby, “innovation within the context of this course meant reaching a viable solution, easy to implement, and that met restrictions such as human resources, monetary funds, and time. Perhaps the level of creativity (a pre-innovation step) was somewhat limited by the source of information (mainly formal research databases). However, students still needed to think differently to accommodate so many constraints”.

The ambiguity of the terms *creativity* and *innovation* left some students unsure if they had achieved an innovative solution to the industry problem unless they received external validation from the industry partner that, “Your idea is innovative.” Other students felt their solutions were not inherently innovative if they were solving a problem in response to a brief if they “did not have to innovate but to solve the problem they had.” These students were not clear about the link between problem-solving, creativity, and innova-

tion in their projects. Students indicated that they wanted more guidance to understand what creativity and innovation meant in the disciplinary context. As one student suggested, this could be achieved by a greater focus on building this understanding in class: “It would be cool if. . . in the class, they said more things about innovation.”

4.5. Theme 5: The perception of how and how effectively teams function varies between the participant-stakeholders

This theme reflects the variability in how teams operated in terms of their decision-making processes, role allocation, communication and conflict resolution, and how effectively they functioned in the eyes of students, lecturers and industry partners. While guidance was given to students on team formation, each team operated quite differently, indicating gaps between lecturer assumptions about the effectiveness of the guidance provided and the reality of team cohesiveness and functioning.

A review of the data indicates that the teamwork aspect of the project was valued and highly beneficial to the students, particularly in a university environment where usually, “everything is very competitive.” In this competitive environment, a student stated that they often “do more for the grade than because [they] want to learn or do something.” In contrast, a substantial benefit of the teamwork approach and the sharing of roles within teams was that it built a level of collaboration and sharing of information and ideas that was very different from the competitive approach to learning in other courses in the engineering program.

A lecturer stated that “the first step in developing the course was building teams.” Resources were provided to students to assist them in forming teams with “a balanced mix” of personalities. Some teams found the resources provided to them were useful in identifying and allocating roles for the effective functioning of a team; however, other teams disregarded resources for team-formation and determined how their team would function by other methods. Where the resources were used, a member of one team stated that “if it had not been part of the required tasks, we would not have done it.” This same student felt that doing the team-building exercise was useful: “What it showed us was a way of organizing, that someone must take care of the laboratories, the documents and how to create a more orderly group.” In groups where there was a lack of defined roles, sometimes things slipped between the cracks, such as booking of laboratories: “We had to reserve the laboratory days before and sometimes we forgot.”

The industry partner observed that the functionality of teams was linked to communication within teams. This was visible in the differences in functionality between the teams the industry partner worked with.

One of them interacted with each other, and they all knew the subject. While each of them presented their slides and their specific part in the project, each complemented the other. With the other team, it did not feel that there was any integration between them, and they did not know [the subject] either.

Decision making approaches also varied between teams. From the lecturers’ perspective, the project teams functioned well, and with no problems and tensions, however, students identified that problems did occur. Teams applied different strategies to resolve issues, in one case imposing financial sanctions that required students to make a payment of some sort, such as to bring food to share with others, if they did not meet their obligations to the group.

4.6. Translating the themes into a course model

The data analysis process culminated in linking the themes into a visual representation of the course (Fig. 2). The model shows the

relationships between the main people involved in the course: the students, the student teams, the lecturers and industry partners, and the problem-solving processes that take place. The employability skills of creativity and team work are visible in the model, as are the constraints that bound the course. The model illustrates the integration of the components of this non-placement work-integrated learning course.

5. Discussion

This study set out to evaluate the participants' perceptions concerning whether the employability skills of creativity and teamwork are developed in the course *Chemical Engineering Challenge*. The findings of the study are contained in five themes presented in the previous section. They indicate that work-integrated learning using the PO/PBL model, combined with the Integrated Product and Project Design (IPPD) tool and supported by lecturers and industry partners to solve authentic problems create a guided learning environment that motivated and inspired student learning and connected students to their future careers as chemical engineers. While the employability skills of creativity and teamwork were built into the problem-solving process, the perceptions of students, lecturers, and the industry partner varied as to the extent to which these were developed. Thus, while the study highlights the strengths of the design and implementation of the course, it also points to constraints and areas for improvement, particularly in student preparation for the challenge and in enhancing the explicitness of teaching and assessing the employability skills of creativity and teamwork.

The strengths of the course are linked to the integrated nature of the course design, as seen in the course model. One reflection of this is the high levels of engagement and motivation of students, stimulated by engagement in authentic tasks with industry and the development of a "pre-professional identity" (Jackson, 2017) as a chemical engineer. The findings of this work show that, when students are treated as employees and have real-world expectations placed on them, they rise to the challenge. This is particularly evident when there is the possibility that student work is adopted into the industry's operations (Johns-Boast and Patch, 2010). A further strength of the course is that it is designed as "non-placement WIL" (Jackson, 2017); that is, it links students with industry projects but does not involve placement in the workplace. As a non-placement WIL course, the course is not impacted by the "imbalance in the supply and demand of placement WIL opportunities and issues with resourcing" (Jackson, 2017, p. 835), nor by industry exploitation of students as a source of cheap labor (Mutereko and Wedekind, 2016). The viability of non-placement WIL in the course is evident at the time of writing, during the SARS-CoV-2 pandemic. While the course has been impacted by lack of access to laboratories due to social distancing requirements, authentic industry projects continued in 2020, albeit with some adjustments.

The study identified some weaknesses and areas for improvement in the course. The findings show that there was a gap between lecturer and student understandings of creativity and innovation in chemical engineering. Lecturers introduced the IPPD tool at a theoretical level during the first weeks of the course, and the lecturers assumed that students would piece together the link between creativity and innovation and the problem-solving process. However, this was not the case, and many students did not see the process of problem-solving or their final products as being creative. This suggests that introducing and using the IPPD tool is not enough. Drawing on the literature, the authors propose that explicit connections need to be made between problem-solving and the creative process (Lopez-Malo et al., 2016; Schmidt and Charney, 2018). Approaches to explicitly assess creativity that have worked

in other higher education contexts include the use of well-defined rubrics to define performance standards of creativity and their use in student self-assessment, to clarify desired performance standards and assist students in building reflective practice (Felder and Brent, 2010; Lopez-Malo et al., 2016).

Another weakness identified by the study was stakeholders' perceptions of differences in teams' effectiveness, which is of concern given that teamwork is one of the skills most sought after by employers (Quacquarelli Symonds, 2018). Despite the use of activities and resources to guide normative teamwork processes, the findings show that the models and frameworks designed for students to use in team formation, role allocation and conflict resolution were only used by some of the teams and some of the time. If and how they were used was not visible to lecturers. The differences in the ways the teams chose to work can be explained in terms of group norms, "the traditions, behavioral standards and unwritten rules that govern" (Duhigg, 2016, para. 19) team dynamics.

While the teams functioned effectively in so much as they were all able to present a solution to an industry problem, the industry partner identified variability in the nature of communication observed in the different teams and noted that this impacted on the quality of their solutions. The authors hypothesize that the nature of communication amongst team members was a reflection of group norms and their influence on team members' psychological safety, that is, their "sense of confidence that the team will not embarrass, reject, or punish someone for speaking up" (Edmondson, 1999, p. 354). This hypothesis is supported by research on teamwork by Google, which found that "what distinguished the 'good' teams from the dysfunctional groups was how teammates treated one another" (Duhigg, 2016, para. 26). That extensive study found that "the right norms... could raise a group's collective intelligence, whereas the wrong norms could hobble a team, even if, individually, all the members were exceptionally bright" (Duhigg, 2016, para. 26). The communication between the industry partner and the teams was also crucial to the project outcomes. The findings of the study indicate that the level of involvement between the industry partner and the teams impacted student motivation and engagement in their project.

Where high levels of psychological safety exist within teams, this "facilitates the willing contribution of ideas and actions to a shared enterprise" (Edmondson and Lei, 2014, p. 24). That is, there is a link between psychological safety and learning that is characterized by "mutual respect and trust among team members" (Edmondson, 1999, p. 354). To create an environment that supports psychological safety as well as team productivity requires many of the structural elements that are already contained in the course, such as "a clear compelling team goal, [and] an enabling team design" (Edmondson, 1999, p. 356). Additionally, effective teamwork requires supportive and non-judgmental behaviors of team members towards one another, modeled by the team leader: "If the leader is supportive, coaching-oriented, and has non-defensive responses to questions and challenges, members are likely to conclude that the team constitutes a safe environment" (Edmondson, 1999, p. 356). This aligns with the notion of high-performance teams which demonstrate "interdependence and trust between members" (Munro and Laiken, 2003, p. 2).

The assessment of employability skills is contentious and difficult (Ferns and Zegwaard, 2014; Ramberg et al., 2019), and it is known that teamwork, specifically, "is difficult to quantify, as it must be inferred from myriad interrelated behaviors and attitudes" (Britton et al., 2017, p. 2). However, preliminary research indicates "that teamwork skills improve over time when taught and assessed" (Britton et al., 2017, p. 1). Teaching teamwork involves establishing and communicating the desired group norms for team interaction (Duhigg, 2016), including those that build trust and respect and developing tools for measuring achievement. Tools

Table 2

Key learnings from the Chemical Engineering Challenge about non-placement WIL using a PO/PBL methodology.

Focus of Learnings	Key Learnings
Student motivation and engagement	- Utilize industry-linked authentic problems. - If possible, partner with prestigious industries. - Discuss with students their commitment to the industry partner (their rights and responsibilities as if they were “employees”).
Guidance and support	- Be explicit about the process of creation and innovation in accordance with the chosen approach (in this case, IPPD). - Establish regular meetings between lecturers and the whole class and individual teams. - Be explicit with students about expectations (e.g., quality of the final product, the time needed, communication with industry). - Negotiate with industry partners on the nature and scope of the challenges to be used in the course.
Individual perceptions of preparedness	- Manage stakeholder expectations (e.g., define the final product). - Mediate between industry partners and students (e.g., amount of work, expectations, time, final product). - Increase the length of the course to a full semester (16 weeks).
Concepts and definitions	- Be explicit with students about the concepts and definitions that are used and how students can apply them. - Introduce the tools and models being used early in the course and refer back to them frequently. - Design class activities that allow students to reflect on their creative process. (e.g., describe your methodology, compare it with the rubrics, identify unique elements of your work). - Build a shared understanding with students on what innovation and creativity mean in relation to the IPPD (or another design tool). - Use well-defined rubrics to evaluate performance standards of creativity and teamwork directly. - Incorporate student self- assessment tasks and rubrics to monitor student and team progress.
Team effectiveness	- Review the expectations of the different stakeholders about the time needed, deadlines and communication. - Incorporate activities for team building and establishing team-norms. - Establish strategies and protocols that students can use to achieve a shared understanding of what good teamwork looks like. - Provide students with resources (such as rubrics and checklists) that allow them to monitor their team's effectiveness. - Design class activities to check-in with students about how their teams are functioning.

such as rubrics may be useful for self and peer reflection of teamwork, as well as formal evaluation of the extent to which those desired behaviors and norms are achieved (Britton et al., 2017).

A final tension identified in the course was the different expectations of students, lecturers, and industry partners about student preparedness, which the authors have shown as ‘constraints’ in Fig. 2. The challenge of coordinating the expectations of various stakeholders in project-based learning courses is not new (Patrick et al., 2008). In this study, the different timeframes and speeds of academia and industry played out as a central point of contention. Industry moves faster than academia, and the challenges it faces require effective and efficient solutions in the short and long term (Cooper et al., 2011). This raises the paradox of students being motivated to learn by engaging with industry-based problems but not having sufficient disciplinary skills and knowledge to do so until towards the end of their degrees. The course was designed to be 8 weeks long. In response to feedback from students in relation to its short duration, the course was extended to 16 weeks in semester 1 of 2020. An additional solution may be to introduce the IPPD tool in other courses in the earlier years of the chemical engineering program, with progressive increases in task authenticity and proximity to the industry as students’ progress through the program (Cooper et al., 2011). This would introduce students to the problem-solving methodology required for engagement in authentic industry-based tasks progressively, thus preparing students with the skills and knowledge needed for the course before enrolling in it.

In summary, the course provides a supportive environment for the development of students’ problem-solving, teamwork, and creativity skills, though one that can be enhanced through some strategic changes in teaching and assessment: the earlier introduction of the IPPD tool, building a shared understandings of effective teamwork, articulating different time perceptions, and acknowledging students skills and preparedness for the challenges. The great strength of the course is the use of the IPPD problem-solving tool to scaffold the problem-solving process: identifying an opportunity, generation, and selection of ideas, prototyping, developing a model and a final solution. A fundamental aspect of the course is how different stakeholders (students, lecturers and industry

partners) are intertwined in the IPPD process, which enriches the project-based learning experience, links university learning with the industry context, and reduces the risk of poor-quality student learning experiences during the project as a result of variable industry contexts. Lecturers guide students to engage professionally with industry partners, gaining first-hand work experience, and experiencing the challenges of their future jobs (Johns-Boast and Patch, 2010). This non-placement approach to WIL provides a transition between a purely academic learning environment and full immersion in the industry where lecturers serve as a translator between industry requirements, the learning outcomes, and the students’ skills and knowledge (Martin and Hughes, 2009).

Drawing from the findings of this project, the authors conclude this discussion with a summary of key learnings that are relevant for the authors in the iterative development of the course (Table 2). These learnings may also be useful for academics in other contexts who are considering the use of non-placement WIL using a Problem-Oriented/Project-Based Learning methodology while recognizing that the course was designed for a specific context and with a particular focus on the employability skills of creativity and teamwork.

6. Conclusion

The integration of technical skills and knowledge with employability skills in chemical engineering programs is essential to prepare work-ready graduates. This paper presented student, lecturer, and industry perceptions of the integration of the employability skills of creativity and teamwork through a non-placement WIL experience in a chemical engineering course at a Colombian university. The problem-oriented/ project-based learning methodology used in the course is relatively new to the university. This study supports its more comprehensive implementation in courses in the faculty to scaffold employability skills’ development as students’ progress through the chemical engineering program and to help them develop their pre-professional identities.

The development of employability skills of creativity and teamwork are built into the course through the use of the IPPD

problem-solving tool. However, the study found that explicit teaching, reflection on, and assessment of these employability skills are required to develop a shared understanding of these concepts and of the behaviors and attitudes that characterize them. Explicit teaching is also needed to assist students in making the connection between creativity and innovation as elements of problem-solving and to see the scaffolding of creativity through the IPPD framework. While assessment of employability skills is problematic, the use of rubrics for self-reflection is one method that could be used to “extend students as active and agentic learners [which] is central to the effective integration of experiences across practice, and higher education settings, their ability to engage in professional practice and their becoming effective critical and reflexive practitioners” (Billett, 2011, p. 1).

In conclusion, the PO/PBL approach taken in the *Chemical Engineering Challenge*, combined with the IPPD framework, provided a blend of guidance and monitoring of students and their progress by their lecturers while facilitating an industry perspective and context for their project from an industry partner. While some adjustment was identified to enhance the development of employability skills in the course, the model of non-placement WIL provided a transitional space for students to engage with the world of work.

Declaration of Competing Interest

The authors report no declarations of interest.

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